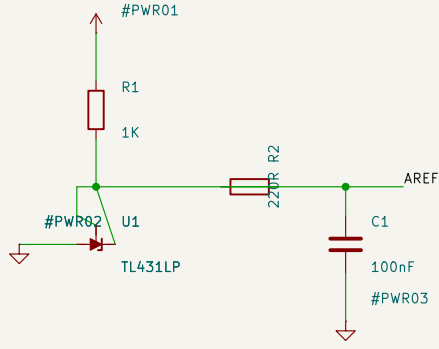
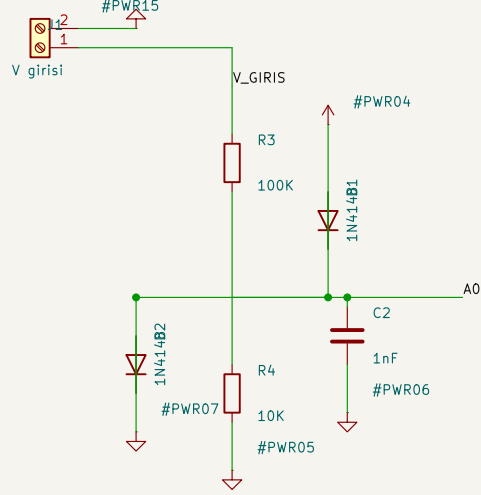


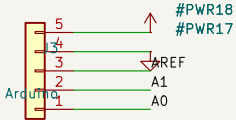
BLOK 1 – TL431 gerilim referansi → AREF



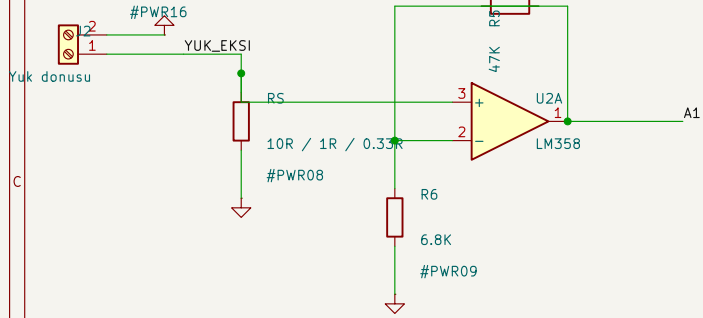
BLOK 2 – Gerilim bolucu 1:11 + koruma kelepçeleri → A0



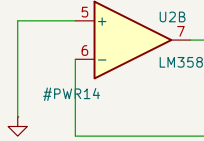
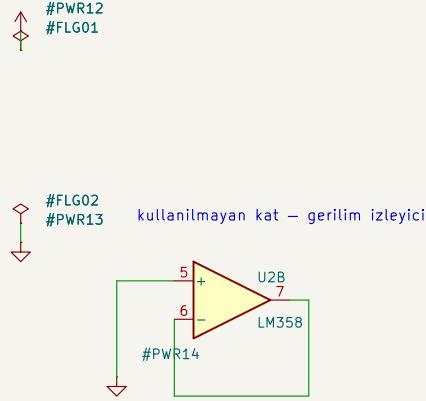
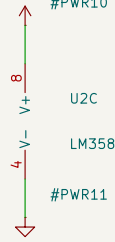
Arduino



BLOK 3 – Sont + LM358 yükselteci (x7.912) → A1



BLOK 4 – Besleme ve baglanti



Tüm değerler üretim/sim_*.py ile doğrulandı. AREF = 2.478 V, tam ölçek 27.2 V / 31–940 mA

UYARI: firmware’de analogReference(EXTERNAL) ilk analogRead’den ÖNCE çağrılmalı – yoksa cip yanar.

R2 = 220R seçildi: AREF iç yükü (~32k) üzerinden düşüm 17 mV. 4.7K olsaydı 320 mV düşerdi.

C2 = 1nF seçildi: kesim 17.8 kHz. 100nF olsaydı 178 Hz olur, osiloskop kipi olurdu.